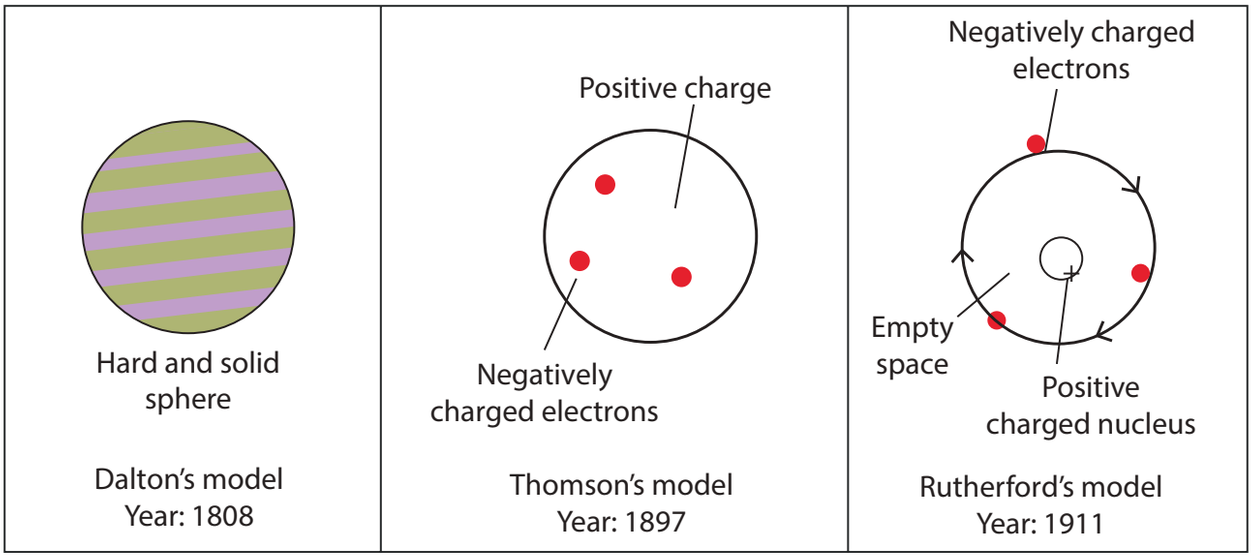
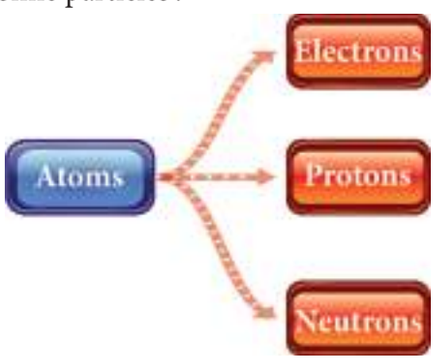
Stages of discovery of the consitituents of an atom



You have around 7 billion atoms in your body, yet you replace about 98% of them every year!

4.2 The sub-atomic particles

The discoveries made during the twentieth century proved that atoms of all elements are made up of smaller components - electron, proton and neutron. An electron from hydrogen atom is no different from the electron of a carbon atom. In the same manner, protons and neutrons of all elements also have same characteristics. These particles that make up the atom are called 'subatomic particles'.



Proton (p)

The proton is the positively charged particle and it is located at the nucleus. Its positive charge is of the same magnitude as that of the electron's negative charge.

Neutron (n)

Neutron is inside the nucleus. The neutron does not have any charge. Except hydrogen (protium), the nucleus of all atoms contain neutrons. Protons and neutrons are the two types of particles in the nucleus of an atom. They are called nucleons.

Electron (e)

This is a negatively charged particle. Electrons revolve around the nucleus of the atom in specific orbits. The mass of an electron is negligible as compared to that of a proton or neutron. Hence, the mass of an atom depends on the number of protons and neutrons in the nucleus.

The total negative charge of all the electrons outside the nucleus is equal to the total positive charge in the nucleus. That makes the atom electrically neutral.

VII Std Science Term-1 EM Unit 4.indd 45 23-01-2025 12.46.21 PM